Note on Commutative Algebra

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Contents

Acknowledgement

1. This note is basically a copy of Eisenbud's Commutative Algebra with a View Toward Algebraic Geometry, with some additional explanations and examples from Eisenbud's lectures and my own understanding.

2. This latex style is made by Evan Chen.

3. In this note, all rings R are commutative with unity unless otherwise specified.

4. When we use the language of category theory, we are in R-modules categories unless otherwise specified.

January 17, 2026: A theorem in Invariant Theory

Corollary 0.1 (Summand of polynomial rings over fields is finitely generated)
Let k be a field, and let R be a k-subalgebra of $S = k[x_1, \dots, x_n]$. R is a direct summand of S in the sense that there is a surjective R-algebra homomorphism $\phi : S \to R$ such that $\phi|_R = \mathrm{id}_R$ and ϕ preserves degrees. Then R is finitely generated as a k-algebra.

Proof. Intuition: R itself is a graded k-algebra, with grading inherited from S. As a graded algebra, R can be decomposed into its homogeneous components. R_0 is easy to get, just include 1 in the generators. Therefore, to generate R, we really need to generate the positive degree part. That motivates us to look at the ideal R_+.

But R_+ may not be finitely generated as an ideal. However, we know that S is a Noetherian ring, so we can look at the ideal R_+S. It must be finitely generated, say by $f_1, \dots, f_m \in R_+$.

Now, we claim that $R = k[1, f_1, \dots, f_m]$. Certainly, the right hand side is contained in the left hand side. For the other direction, it suffices to show that any homogeneous element of R can be generated by $f_1, \dots, f_m$ with coefficients in k. We prove this by induction on degree d. $d = 0$ is trivial. Now suppose we have shown this for degree less than d. Take any homogeneous element $r \in R$ of degree d. Since $r \in R_+S$, we can write $r = \sum g_i f_i$ for some homogeneous element $g_i \in S$. Apply ϕ to both sides, we get

$$r = \phi(r) = \sum \phi(g_i)\phi(f_i) = \sum \phi(g_i) f_i.$$

Note that $\deg(\phi(g_i)) = \deg(g_i) < d$, so by the inductive hypothesis, each $\phi(g_i)$ can be generated by $f_1, \dots, f_m$ over k. Therefore, r can also be generated by $f_1, \dots, f_m$ over k. This finishes the proof. $\square$

Remark 0.2. This shows that the invariant ring of a linearly reductive group acting on a polynomial ring over a field is finitely generated. If the group G is finite, the surjective homomorphism ϕ can be taken as $\phi(f) = \sum_{g \in G} gf$. (Weyl's averaging trick)

Remark 0.3. Graded rings are great because it allows us to use induction on degree. So we only need to worry about homogeneous elements.

January 20, 2026: Basis, syzygy, and Hilbert's function theorems

First day of Commutative Algebra course!

We discuss the history of Commutative Algebra and three of the four big theorems.

Theorem 0.4 (Hilbert's Basis Theorem)
If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.

Proof. Let I be an ideal of $R[x]$. Our goal is to show that I is finitely generated. We first pick the smallest degree nonzero polynomial f_1 in I. Next, we take f_2 to be the smallest degree polynomial in I which is not in the ideal generated by f_1, that is $f_2 \in I \setminus (f_1)$. Continuing this process, we get a sequence of polynomials $f_1, f_2, \ldots$ in I such that f_n is the smallest degree polynomial in $I \setminus (f_1, \ldots, f_{n-1})$. Clearly we have $\deg(f_{i+1}) \geq \deg(f_i)$ for all i. Since R is Noetherian, the leading coefficients of f_i generate a finitely generated ideal in R. This means that there exists j such that the leading coefficient of f_1 to f_j generate all the leading coefficients of f_i for $i > j$.

Now, we look at f_{j+1}. Its leading coefficient can be generated by the leading coefficients of f_1 to f_j, that is $a_{j+1} = \sum_{i=1}^{j} r_i a_i$ for some $r_i \in R$. Consider the polynomial

$$g = f_{j+1} - \sum_{i=1}^{j} r_i x^{\deg(f_{j+1}) - \deg(f_i)} f_i$$

Since $\deg(g) < \deg(f_{j+1})$ so $g \in (f_1, \ldots, f_j)$. Thus, $f_{j+1} \in (f_1, \ldots, f_j)$, contradicting the choice of f_{j+1}. Therefore, $I = (f_1, \ldots, f_j)$ is finitely generated. $\square$

Theorem 0.5 (Hilbert's syzygy theorem)
If $R = k[x_1, \ldots, x_n]$ is a graded polynomial ring over a field k, then every finitely generated graded R-module $M = \bigoplus_{s \in \mathbb{Z}} M_s$ has a finite graded free resolution of length at most n. That is, the sequence

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0,$$

where each F_i is a finitely generated graded free R-module, is exact.

Proof is omitted.

Theorem 0.6 (Hilber's function theorem)

Let M be a finitely generated graded module over the polynomial ring $R = k[x_1, \ldots, x_n]$. Then there exists a polynomial $P(t) \in \mathbb{Q}[t]$ such that for all sufficiently large d, we have $\dim_k M_d = H_M(d) = P(d)$.

We will present two proofs of this theorem.

Proof. We will use the Hilbert's syzygy theorem to prove this. By the theorem, we have a finite graded free resolution of M:

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0.$$

Thus, we have $H_M(d) = \sum_{i=0}^{n} (-1)^i H_{F_i}(d)$. So we only need to study the dimension of free modules F_i, which is a direct sum of $k[x_1, \ldots, x_n]$. Note that $H_R(d) = \dim_k R_d = \binom{n+d-1}{n-1}$ (star and bar counting trick), which is a polynomial in d when d is sufficiently large. Therefore, $H_M(d)$ is also a polynomial in d for sufficiently large d. $\square$

Remark 0.7 (The Euler characteristic)**.** Why does $H_M(d) = \sum_{i=0}^{n} (-1)^i H_{F_i}(d)$ hold? This result is a property of the Euler characteristic. For any complex of finite-dimensional vector spaces

$$V_* : 0 \to V_n \to V_{n-1} \to \cdots \to V_1 \to V_0 \to 0,$$

We define its Euler characteristic to be

$$\chi(V_*) = \sum_{i=0}^{n} (-1)^i \dim_k V_i.$$

There is a nontrivial fact that the Euler characteristic is 0 if the complex is exact. The easiest proof is via the decomposition of exact sequence into short exact sequences and sum up the dimension relations for each short exact sequence.

$$\begin{tikzcd}
 & & 0 \arrow[dr] & & 0 & & \\
 & & & \operatorname{im} d_{i+1} = \ker d_i \arrow[ur] \arrow[dr] & & & \\
M_{i+2} \arrow[rr, "d_{i+2}"] \arrow[dr] & & M_{i+1} \arrow[rr, "d_{i+1}"'] \arrow[ur] & & M_i \arrow[rr, "d_i"] \arrow[dr] & & M_{i-1} \\
 & \operatorname{im} d_{i+2} = \ker d_{i+1} \arrow[ur] \arrow[dr] & & & & \operatorname{im} d_i = \ker d_{i-1} \arrow[ur] \arrow[dr] & \\
0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0
\end{tikzcd}$$

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Second one is via induction on n.

Proof. $n = 0$ is trivial. Suppose it's true for number less than n. Consider the short exact sequence

$$0 \to (0 : x_n) \to M(-1) \xrightarrow{x_n} M \to M/x_nM \to 0.$$

Notice that M/x_nM is a finitely generated graded module over $k[x_1, \ldots, x_{n-1}]$ (x_n acts as zero) and $(0 : x_n)$ is also a finitely generated graded module over $k[x_1, \ldots, x_{n-1}]$ (the generators of $(0 : x_n)$ over $k[x_1, \ldots, x_{n-1}, x_n]$ also generate $(0 : x_n)$ over $k[x_1, \ldots, x_{n-1}]$ since x_n would act as zero). By the inductive hypothesis, both M/x_nM and $(0 : x_n)$ have Hilbert dimensions agree with polynomials when d is large enough. Thus, by the exactness of the sequence,

$$H_M(d) - H_M(d-1) = H_{M/x_nM}(d) - H_{(0:x_n)}(d-1) = \lambda(d).$$

for some polynomial $\lambda(d)$ when d is large enough. Thus,

$$H_M(d) - H_M(0) = \sum_{i=1}^{d}(H_M(i) - H_M(i-1)) = \sum_{i=1}^{d} \lambda(i).$$

Notice that $\sum_{i=1}^{d} i^k$ is a polynomial in d of degree $k+1$. Therefore, $\sum_{i=1}^{d} \lambda(i)$ is also a polynomial in d when d is large enough. Therefore, $H_M(d)$ also agrees with a polynomial when d is large enough. □

Remark 0.8. Is $H_M(d)$ always finite? The answer is yes if M is a finitely generated module over $k[x_1, \ldots, x_n]$. This is nontrivial because $H_M(d)$ is considering the dimension of M as module over k, not over R.

Proof. Consider $M' = \bigoplus_{s=d}^{\infty} M_s$. M' is a submodule of M, so it's finitely generated as R-module. Let $m_1, \ldots, m_t$ be the generators of M'. Each m_i has degree at least d. So for any element in M_d, it's a linear combination of m_i with coefficients in R of degree at most 0 (otherwise it would have degree greater than d). Therefore, M_d is spanned by $m_1, \ldots, m_t$ over k, which is finite-dimensional. □

I came up with another more intuitive proof.

Proof. M is finitely generated as R-module by $m_1, \ldots, m_t$. For some degree d, any element in M_d must be a linear combination of m_i with coefficients as the monomials in R of degree at most $d - \deg(m_i)$. There are only finitely many such monomials, so M_d is finite-dimensional over k. □

January 22, 2026: Nullstellensatz and equivalence of algebraic varieties and affine algebras

Today we discuss the Hilbert's Nullstellensatz and the equivalence between the category of affine algebraic varieties and the category of finitely generated reduced k-algebras. For further details, please refer to here.

January 27, 2026: Localization and Tensor Products

There is a readiness exam today.

For problem 3, we have a stronger result. For any R and ideals I and J, we have an isomorphism

$$\frac{R}{I} \otimes_R \frac{R}{J} \cong \frac{R}{I+J}.$$

Proof. We will show a more general result: For any ideal I, we have the exact sequence

$$0 \to I \to R \to R/I \to 0.$$

Tensoring with N, we get the right exact sequence

$$I \otimes_R N \to R \otimes_R N \to (R/I) \otimes_R N \to 0.$$

Notice that the image of $I \otimes_R N$ in $R \otimes_R N \cong N$ is exactly IN. Thus, we have the isomorphism $\frac{N}{IN} \cong (R/I) \otimes_R N$. Taking $N = R/J$, we get

$$\frac{R/J}{I(R/J)} \cong (R/I) \otimes_R (R/J).$$

Note that $I(R/J) = IR/J = (I+J)/J$ (an elementary result, informally $I(R/J)$ is just I/J, but I may not contain J so we just use $I + J/J$ to avoid confusion), so we have

$$\frac{R/J}{I(R/J)} \cong \frac{R}{I+J}.$$

Combining the two isomorphisms, we get the desired result. □

Next, we discuss localization. We essentially invert elements of a multiplicative subset $U \subseteq R$. The key propositions are the following:

1. There is a natural map $\phi : R \to M[U^{-1}]$ given by $r \mapsto \frac{r}{1}$.
2. For any ideal $J \subseteq R[U^{-1}]$, $I^c = \phi^{-1}(I)$ is an ideal of R disjoint from U. (This is true in general for any ring homomorphism ϕ.)
3. For any ideal $I \subseteq R$ disjoint from U, the extension $I^e = U^{-1}I$ is a proper ideal of $R[U^{-1}]$. (if they are not disjoint, then 1 is in the $U^{-1}I$, so it's not proper).

4. For any $J \subseteq R[U^{-1}]$, we have $J = (J^c)^e$.

5. However, the $(I^e)^c = I$ is false, even if $I \cap S = \emptyset$. Example: let $U = \{1, x, x^2, \ldots\}$ and $R = \mathbb{C}[x, y]$, $(xy)^e = (y)$ in $R[U^{-1}]$ and $((xy)^e)^c = (y)$ in R. The correct statement is
$$(I^e)^c = \{r \in R : \exists u \in U, ur \in I\} = \sum_{u \in U} (I : u) \supseteq (I : U) \supseteq I$$
. If I is prime, then $(I^e)^c = I$.

6. This gives us the following bijection of prime ideals between R and $R[U^{-1}]$:

$$\begin{tikzcd}
\{\mathfrak{q} \subseteq R[U^{-1}]\} \arrow[rr, bend left, "\phi^{-1}(-)"] & & \{\mathfrak{p} \subseteq R : \mathfrak{p} \cap U = \emptyset\} \arrow[ll, bend left, "U^{-1}(-)"]
\end{tikzcd}$$

Other properties of localization that I think are important:

1. The universal property of localization: for any R-module M and N, multiplicative closed set U. If U acts invertibly on N, then any R-module homomorphism $\psi : M \to N$ factors uniquely through $U^{-1}M$. That is, there exists a unique R-module homomorphism $\tilde{\psi} : U^{-1}M \to N$ such that the following diagram commutes:

$$\begin{tikzcd}
M \arrow[rr, "\psi"] \arrow[dr, "\phi"'] & & N \\
& U^{-1}M \arrow[ur, "\tilde{\psi}"'] &
\end{tikzcd}$$

2. If R is Noetherian, then so is $R[U^{-1}]$. If R is UFD, then so is $R[U^{-1}]$.

3. Localization commutes with direct sums and direct limits because it's a left adjoint functor. It's left adjoint to the forgetful functor from $R[U^{-1}]$-modules to R-modules, forgetting the inverted elements' action.

4. Left adjointness implies that localization is right exact. In fact, localization is an exact functor.

5. Exactness implies that localization preserves finite limits and finite colimits. In particular, it preserves kernels, cokernels (quotient of modules), finite intersections (intersetctions= a fiber product=a finite limit) and finite products. Left adjointness implies that it preserves colimits (need not to assume finiteness). In general, left exactness preserves finite limits, right exactness preserves finite colimits.

Proof. Exactness implies preservation split exact sequences. Thus, this fucntor preserves products and kernels. Any finite limit can be constructed using products and kernels, so it preserves finite limits. Colimit is similar. □

6. Localization is nice with respect to tensor products: Let M and A be R-modules, N be an $R[U^{-1}]$-module, then we have the isomorphism
$$U^{-1}R \otimes_R M \cong U^{-1}M,$$
$$U^{-1}(M \otimes_R N) \cong U^{-1}M \otimes_{U^{-1}R} N,$$
$$U^{-1}(A \otimes_R M) = U^{-1}A \otimes_{U^{-1}R} U^{-1}M.$$
The first isomorphism tells us $(-) \otimes_R U^{-1}R$ is exact, that is $U^{-1}R$ is a flat R-module.

7. Tensor products are associative: for any R-module M, R-S-bimodule N and S-module P, we have the isomorphism
$$(M \otimes_R N) \otimes_S P \cong M \otimes_R (N \otimes_S P).$$

We also discuss some property of tensor products and Hom functors.

1. $(-) \otimes_R M$ is left adjoint to $\mathrm{Hom}_R(M, -)$. Thus, $\mathrm{Hom}_R(M, -)$ is right exact.

2. $h_N(-) = \mathrm{Hom}_R(-, N) : \text{R-Mod}^{\mathrm{op}} \to \text{R-Mod}$ is right adjoint to $h_N(-)^{\mathrm{op}} : \text{R-Mod} \to \text{R-Mod}^{\mathrm{op}}$.

January 29, 2026: More on Localization

Some multiplicative closed sets that are often used:

1. $U = R_x = 1, x, x^2, \ldots$ for some $x \in R$. The localization R_x is often denoted as $R[x^{-1}]$.

2. $U = R \setminus \mathfrak{p}$ for some prime ideal $\mathfrak{p}$. $U^{-1}R$ is often denoted as $R_{\mathfrak{p}}$ and called the localization of R at $\mathfrak{p}$.

Example 0.9 (The Ring of Laurent Polynomials)
$k[x]_x = k[x, x^{-1}]$.

Example 0.10
The regular function on an affine open subset $D(f)$ of an affine variety X is given by the localization of the coordinate ring $A(X)$ at f, that is $A(X)_f$.

Example 0.11 (stalk of a variety at a point)

Let $X \subseteq \mathbb{A}^n$ be an affine variety over an algebraically closed field k. For any point $p \in X$, the stalk of X at p (all regular functions that are well-defined locally at p) is isomorphic to the localization of the coordinate ring $A(X)$ at the maximal ideal $\mathfrak{m}_p$ corresponding to the point p. That is,

$$\mathcal{O}_{X,p} = A(X)_{\mathfrak{m}_p} = \{\frac{f}{g} \mid f, g \in A(X), g(p) \neq 0\}.$$

Example 0.12

$(\mathbb{Z}/10\mathbb{Z})_2 \cong \mathbb{Z}/5\mathbb{Z}$.

Proof. $(\mathbb{Z}/(10))_2 = (\mathbb{Z}_2/(10)_2) = (\mathbb{Z}_2/(5)_2) = (\mathbb{Z}/(5))_2 = \mathbb{Z}/5\mathbb{Z}$.
Most equalities are guaranteed by the fact that localization preserves quotients. The last step is because $\mathbb{Z}/(5)$ is a field, so any nonzero element is invertible. □

Theorem 0.13 (Local Global Principle)

$M = 0$ if and only if $M_\mathfrak{p} = 0$ for all prime ideals $\mathfrak{p}$ if and only if $M_\mathfrak{m} = 0$ for all maximal ideals $\mathfrak{m}$.

Remark 0.14. One way to understand this theorem is through dimension argument in algebraic geometry. The codimension of a point (the largest dimension of objects containing the point) is the same as the height of the corresponding prime ideal, which is the Krull dimension of the localized ring at that prime ideal. If the codimenson of any point is 0, then the variety is empty! So the coordinate ring of the variety is 0.

Another way to understand this theorem is through the stalks of a sheaf. If a ring of regular functions is 0 at every stalk (the small neighborhood of every point), then the only function in the ring is the zero function. So the ring itself is 0.

Theorem 0.15

A sequence of R-modules is exact if and only if the localized sequence at any prime ideal is exact if and only if the localized sequence at any maximal ideal is exact.

In particular, an R-module homomorphism $\phi : M \to N$ is injective (surjective) if and only if the localized homomorphism $\phi_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is injective (surjective) for all prime ideals $\mathfrak{p}$ if and only if the localized homomorphism $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective (surjective) for all maximal ideals $\mathfrak{m}$.

Example 0.16 (Chinese Remainder Theorem)

Let R be a ring, and let Q_i be ideals of R such that $Q_i + Q_j = R$ for all $i \neq j$. Then we have the isomorphism

$$R/\bigcap_{i=1}^{n} Q_i \cong \bigoplus_{i=1}^{n} R/Q_i.$$

Furthermore, one can show that $\bigcap_{i=1}^{n} Q_i = \prod_{i=1}^{n} Q_i$.

Proof. We have a natural injection from $R/\bigcap_{i=1}^{n} Q_i$ to $\bigoplus_{i=1}^{n} R/Q_i$ given by $r \mapsto (r + Q_1, \ldots, r + Q_n)$. We will show that this map is also surjective by localization.
For any maximal ideal $\mathfrak{m}$ of R, there is at most one Q_i contained in $\mathfrak{m}$ since $Q_i + Q_j = R$ for all $i \neq j$. Since Q_j is not contained in $\mathfrak{m}$, then the localization of Q_j at $\mathfrak{m}$ is the whole localized ring $R_{\mathfrak{m}}$.

When we localize the injection at $\mathfrak{m}$, we get the map

$$R_{\mathfrak{m}}/\bigcap_{i=1}^{n} (Q_i)_{\mathfrak{m}} \to \bigoplus_{i=1}^{n} R_{\mathfrak{m}}/(Q_i)_{\mathfrak{m}}.$$

That is,

$$R_{\mathfrak{m}}/(Q_i)_{\mathfrak{m}} \to R_{\mathfrak{m}}/(Q_i)_{\mathfrak{m}}.$$

which is clearly an isomorphism. Thus, the localized map is an isomorphism for any maximal ideal $\mathfrak{m}$. By the local global principle, the original map is also an isomorphism. $\square$

Theorem 0.17

Let R be a ring, and S be a R-algebra and S is flat as an R-module. If M and N are S-modules, and M is finitely presented as an R-module, then we have the isomorphism

$$S \otimes_R \mathrm{Hom}_R(M, N) \cong \mathrm{Hom}_S(M \otimes_R S, N \otimes_R S).$$

Proof. There is a natural homomorphism

$$\alpha_M : S \otimes_R \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_S(M \otimes_R S, N \otimes_R S)$$

sending $s \otimes f$ to the homomorphism $m \otimes s' \mapsto f(m) \otimes ss'$.

First, if M is a free R-module, then α is an isomorphism, this is very easy. Thus, for any finitely presented R-module M, we have an exact sequence

$$F \to G \to M \to 0.$$

where F and G are free R-modules. Applying $\mathrm{hom}_R(-, N)$, we get the exact sequence

$$0 \to \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(G, N) \to \mathrm{Hom}_R(F, N).$$

Applying $S \otimes_R -$ (S is flat), we get the exact sequence

$$0 \to S \otimes_R \mathrm{Hom}_R(M, N) \to S \otimes_R \mathrm{Hom}_R(G, N) \to S \otimes_R \mathrm{Hom}_R(F, N).$$

On the other hand, applying $- \otimes_R S$ to the first exact sequence, we get the exact sequence

$$F \otimes_R S \to G \otimes_R S \to M \otimes_R S \to 0.$$

Applying $\mathrm{Hom}_S(-, N \otimes_R S)$, we get the exact sequence

$$0 \to \mathrm{Hom}_S(M\otimes_R S, N\otimes_R S) \to \mathrm{Hom}_S(G\otimes_R S, N\otimes_R S) \to \mathrm{Hom}_S(F\otimes_R S, N\otimes_R S).$$

We have the following commutative diagram:

$$\begin{tikzcd}
0 \arrow[r, "0"] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r, "0"] \arrow[d] & 0 \arrow[d] \\
S \otimes_R \mathrm{Hom}_R(M, N) \arrow[r, "\alpha_M"] \arrow[d] & \mathrm{Hom}_S(M \otimes_R S, N \otimes_R S) \arrow[d] \\
S \otimes_R \mathrm{Hom}_R(G, N) \arrow[r, "\alpha_G"] \arrow[d] & \mathrm{Hom}_S(G \otimes_R S, N \otimes_R S) \arrow[d] \\
S \otimes_R \mathrm{Hom}_R(F, N) \arrow[r, "\alpha_F"] & \mathrm{Hom}_S(F \otimes_R S, N \otimes_R S)
\end{tikzcd}$$

By five lemma, α_M is an isomorphism.

□

Application 0.18 (Eisenbud's Exercise 2.13)
If C is finitely presented, then the short exact sequence

$$0 \to A \to B \to C \to 0$$

splits if and only if

$$0 \to A_p \to B_p \to C_p \to 0$$

splits for all prime ideals p.

The key idea is to use the facts: R_p is flat and

$$0 \to A \to B \to C \to 0$$

splits if and only if

$$0 \to \hom(C, A) \to \hom(C, B) \to \hom(C, C) \to 0$$

is exact.

Theorem 0.19

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}$$

Proof. One direction is trivial. For the nontrivial one, assume $f \in \mathfrak{p}$ for all prime ideals $\mathfrak{p}$ containing I. We want to show that $f \in \sqrt{I}$. If not, then we can localize at $\{|f^n : n \geq 0\}$. In the localized ring, I_f is a proper ideal, so there exists a maximal ideal $\mathfrak{m}$ containing I_f. $\mathfrak{m}$ corresponds to a prime ideal $\mathfrak{p}$ in the original ring containing I but not containing f, contradicting the assumption. □

February 3, 2026: Finite Length and Artinian

Today we discuss the finite length condition.

Definition 0.20. We define the length of a module M to be the smallest integer $\ell(M) = n$ such that there exists a chain of submodules of M of length n:

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0.$$

If such n does not exist, we say M has infinite length. If M has finite length, we say M is of finite length.

Lemma 0.21

Let N be a simple R-module, then $N \cong R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ of R. In fact, $\mathfrak{m} = \mathrm{Ann}(N)$.

Proposition 0.22

Let R be a ring, and M be an R-module. M has a finite composition series if and only if M is both Noetherian and Artinian. If $M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$ has a finite composition series, then

1. Any strictly descending chain of submodules of M has length at most n.
2. Any composition series of M has length n. This gives a well-defined notion of the length of M, denoted as $\ell(M)$, which is the length of any composition series of M.
3. The sum of localization maps $M \to M_{\mathfrak{m}}$ for all maximal ideals $\mathfrak{m}$ gives an isomorphism of R-modules:
$$M \cong \bigoplus_{\mathfrak{m}} M_{\mathfrak{m}}$$
where the sum is take over all maximal $\mathfrak{m}$ such that $\frac{M_i}{M_{i+1}} \cong R/\mathfrak{m}$ for some i. Note that for each maximal ideal $\mathfrak{m}$, we only use it once even if it appears multiple times as composition factors.
4. The number of composition factors isomorphic to $R/\mathfrak{m}$ is the same as the length of $M_{\mathfrak{m}}$ as $R_{\mathfrak{m}}$-module. Thus, it's indepenent of the choice of composition series. Furthermore, any two composition series of M have the same multiset of composition factors.
5. $M = M_{\mathfrak{m}}$ if and only if M is annihilated by some power of $\mathfrak{m}$.
6. For any submodule $N \subseteq M$, $\ell(M) = \ell(N) + \ell(\frac{M}{N})$. And the composition factors of M is the disjoint union of the composition factors of N and M/N as multisets.

Proof. If M is Artinian and Noetherian, we can pick a maximal proper submodule M_1 of M (by Noetherian condition), then M/M_1 is simple. Then, we pick a maximal proper submodule M_2 of M_1, and so on. We can continue this process to get a composition series. It must be finite because M is Artinian.

First, show that if M' is a proper submodule of M, then $\ell(M') < \ell(M)$, that is for any composition series of M of length n, we can find a composition

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series of M' of length less than n. Just take the intersection of M' with the composition series of M. This would give us $M' \supseteq M' \cap M_1 \supseteq M' \cap M_2 \supseteq \cdots \supseteq M' \cap M_n = 0$. We claim that this is a composition series of M'. Notice that $\frac{M' \cap M_i}{M' \cap M_{i+1}} \cong \frac{M' \cap M_i + M_{i+1}}{M_{i+1}} \subseteq \frac{M_i}{M_{i+1}}$ is either 0 or $\frac{M_i}{M_{i+1}}$ which are btoh simple. And it can't be $\frac{M_i}{M_{i+1}}$: if $M' \cap M_i + M_{i+1} = M_i$ for all i, $M' \supseteq M_i$ for all i (induct from $i = n$ to $i = 0$), so $M' = M$, contradiction. Thus, we have $\ell(M') < \ell(M)$.

Second, we show that for any strictly descending chain of submodules of M, its length is at most $\ell(M)$. We proceed by induction on $\ell(M)$. If $\ell(M) = 0$, trivial. If $\ell(M) = n$, we have $M = M_0 \supsetneq N_1 \supsetneq \cdots \supsetneq N_k = 0$. Notice N_1 is a proper submodule of M, so $\ell(N_1) < \ell(M) = n$. By the inductive hypothesis, $k - 1 \leq \ell(N_1) < n$. Thus, $k \leq n$.

Third, we can show that any composition series of M has length n. For any $M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$ is a composition series of M. By the definiton of $\ell(M)$, we have $\ell(M) \leq n$. On the other hand, by what we just proved, any strictly descending chain of submodules of M has length at most $\ell(M)$, so $n \leq \ell(M)$. Thus, $n = \ell(M)$. So all composition series of M have the same length, which is $\ell(M)$.

Fourth, we can show that M is Noetherian and Artinian. For any ascending chain of submodules of M, its length is at most $\ell(M)$, so it must stabilize. For any descending chain of submodules of M, its length is at most $\ell(M)$, so it must stabilize.

Fifth, we want to show (3). It suffices to show the isomorphism when we localize at any maximal ideal Q. We begin with the case when $\ell(M) = 1$, that is, when M is simple. $M \cong R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. If $\mathfrak{m} = Q$, then $M_Q \cong (\frac{R}{Q})_Q$. Since $\frac{R}{Q}$ is a field, any nonzero element is invertible, so $(\frac{R}{Q})_Q \cong \frac{R}{Q}$. If $\mathfrak{m} \neq Q$, then $M_Q \cong (\frac{R}{\mathfrak{m}})_Q = 0$ since $\mathfrak{m}$ becomes the whole ring after localization. In particular, we conclude that if Q and Q' are two different maximal ideals and M is a simple module, then $(M_Q)_{Q'} = 0$.

Sixth, we now return to the general case. Let M be an R-module of length n. We have a composition series of M:

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0.$$

We can localization at any maximal ideal Q to get the series of M_Q:

$$M_Q = (M_0)_Q \supseteq (M_1)_Q \supseteq \cdots \supseteq (M_n)_Q = 0.$$

Notice that $\frac{(M_i)_Q}{(M_{i+1})_Q} \cong (\frac{M_i}{M_{i+1}})_Q$ is either 0 or $\frac{M_i}{M_{i+1}}$. So this is a composition series of M_Q (some terms may be repeated). We denote $\frac{M_i}{M_{i+1}} \cong R/\mathfrak{m}_i$. If $\mathfrak{m}_i = Q$, then $\frac{(M_i)_Q}{(M_{i+1})_Q} \cong R/Q$. If $\mathfrak{m}_i \neq Q$, then $\frac{(M_i)_Q}{(M_{i+1})_Q} = 0$. Thus, the length of M_Q is the number of composition factors of M isomorphic to R/Q. This tells us the compistion factors of M is indepenent from the choice of composition series, and any two composition series of M have the same multiset of composition factors. In particular, if there is no composition factor of M isomorphic to R/Q, then $M_Q = 0$. Furthermore, $(M_Q)_{Q'} = 0$ for any two distinct maximal ideals Q and Q'.

Seventh, consider $\phi : M \to \bigoplus_{\mathfrak{m}} M_{\mathfrak{m}}$ where $\mathfrak{m}$ runs through all maximal ideals corresponding to the composition factors of M. Since $M_Q = 0$ for any maximal ideal Q that is not corresponding to the composition factors of M, we can harmlessly extend the sum to all maximal ideals of R. We will show that ϕ is an isomorphism.

For any maximal ideal Q, we have

$$\phi_Q : M_Q \to M_Q$$

Since $(M_Q)_{Q'} = 0$ for any two distinct maximal ideals Q and Q', the localization of $\bigoplus_{\mathfrak{m}} M_{\mathfrak{m}}$ at Q is just M_Q. Thus, ϕ_Q is an isomorphism. By the local global principle, ϕ is an isomorphism.

Eighth, we let $M = \oplus_{\mathfrak{m}_i} M_{\mathfrak{m}_i}$. If M is annihilated by some power of $\mathfrak{m}$, then we claim that $M = M_{\mathfrak{m}}$. If not, then there exists some maximal ideal $\mathfrak{m}_j \neq \mathfrak{m}$ such that $M_{\mathfrak{m}'} \neq 0$. Therefore, there exists $a \in \mathfrak{m}$ but $a \notin \mathfrak{m}_j$. Since $M_{\mathfrak{m}_j}$ is annihilated by some power of $\mathfrak{m}$, there exists some n such that $a^n M_{\mathfrak{m}_j} = 0$. On the other hand, since $a \notin \mathfrak{m}_j$, a is invertible in $R_{\mathfrak{m}_j}$, so $M_{\mathfrak{m}_j} = 0$, contradiction. Thus, we have $M = M_{\mathfrak{m}}$. Conversely, if $M = M_{\mathfrak{m}}$, then the compistion factors of M are all isomorphic to $R/\mathfrak{m}$, so M is annihilated by some power of $\mathfrak{m}$.

Lastly, combining the composition series of N and M/N, we can get a composition series of M. Thus, $\ell(M) = \ell(N) + \ell(M/N)$. In particular, if N and M/N are of finite length, then so is M. Conversely, we can always construct a composition series of M containing N to get a compistion series of N and M/N. Therefore, the disjoint union of their compistion factors is the composition factors of M as multisets. □

Remark 0.23. Notice that $M \cong \bigoplus_{\mathfrak{m}} M_{\mathfrak{m}}$. This is direct sum. **NOT** direct product. And the proof doesn't work with infinite direct product because localization is left adjoint, so it preserves colimits but not limits.

Theorem 0.24

Let R be a ring. The following are equivalent:

1. R is of finite length as an R-module.

2. R is Artinian.

3. R is Noetherian and has Krull dimension 0 (that is, all prime ideals of R are maximal).

If that's the case, then there are finitely many prime ideals=maximal ideals of R.

Proof. (1) to (2) is proven by the previous proposition.

(2) to (3):

Claim 0.25 — 0 is the product of finitely many maximal ideals.

Proof. Let $\mathcal{F}$ the be family of all ideals that are product of finitely many maximal ideals (can be repeated). $\mathcal{F}$ is clearly nonempty, thus by Artinian condition, there is a minimal element in $\mathcal{F}$, say J.
By the maximality of J, for any maximal ideal $\mathfrak{m}$, $J\mathfrak{m} = J$ and $J \subseteq M$. We also have $J^2 \subseteq J$, so $J^2 = J$.

We claim that $J = 0$. If not, let $\mathcal{F}'$ be the family of all ideals I such that $IJ \neq 0$. Let I be the minimal element of $\mathcal{F}'$. Thus, $IJ = IJ^2 = (IJ)J$. Thus, $IJ = I$ by the minimality of I. So there must exists $f \in I$ such that $(f)J \neq 0$. So $I = (f)$. So $(f)J = (f)$. Thus, there eixsts $g \in J$ such that $fg = f$. So $f(1-g) = 0$. Since J is contiained in all maximal ideals, $1-g$ is not contained in any maximal ideals, so $1-g$ is invertible. Thus, $f = 0$, contradiction. So $J = 0$. □

Thus, we have $0 = \mathfrak{m}_1 \cdots \mathfrak{m}_n$ for some maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. This implies that for any prime ideal $\mathfrak{p}$, $\mathfrak{p} \subseteq (0) = \mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \mathfrak{m}_i$ for some i. We claim that $\mathfrak{p} \supseteq \mathfrak{m}_i$. If not, there eixsts $f_i \in \mathfrak{m}_i$ but $f_i \notin \mathfrak{p}$ for all i. But $\prod_{i=1}^n f_i \in \mathfrak{m}_1 \cdots \mathfrak{m}_n = 0 \subseteq \mathfrak{p}$, so at least one $f_i \in \mathfrak{p}$. Thus, $\mathfrak{p} \supseteq \mathfrak{m}_i$ is maximal. Therefore, any prime ideal of R is maximal and there are only finitely many prime ideals of R.

We now show that R is Noetherian. Consider $M_i = \frac{\mathfrak{m}_1 \cdots \mathfrak{m}_i}{\mathfrak{m}_1 \cdots \mathfrak{m}_i \mathfrak{m}_{i+1}}$ as $R/\mathfrak{m}_{i+1}$-vector spaces.

Claim 0.26 — M_i is finite-dimensional as $R/\mathfrak{m}_{i+1}$-vector space.

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Proof. For any subspace of M_i, it corresponds to an ideal $I \subseteq R$. Since R is Artinian, any descending chain of subspaces of M_i must stabilize, so M_i is finite-dimensional. Thus, we can construct a $M_i \supseteq M_{i,1} \supseteq \cdots \supseteq M_{i,n_i} = 0$ such that $M_{i,j}/M_{i,j+1} \cong R/\mathfrak{m}_{i+1}$ for all j. This is a finite composition series of M_i as R-module, so M_i is of finite length as R-module. Putting the composition series of M_i together, we can get a composition series of R. Thus, by the previous proposition, R is Noetherian. □

(3) to (1): Assume R is not of finite length. Consider $\mathcal{F}$ to be the family of ideals such that R/I is not of finite length as R-module. $\mathcal{F}$ is clearly nonemtpy since $(0) \in \mathcal{F}$, so there is a maximal element I in $\mathcal{F}$.

Claim 0.27 — I is a prime ideal.

Proof. if not, then there exisst $ab \in I$ but $a, b \notin I$. Thus, $(I : a) \supsetneq I$ and $(I) + (a) \supsetneq I$. Thus, $\frac{R}{(I:a)}$ and $\frac{R}{I+(a)}$ are both of fintie length. Consider the exact sequence of R-modules:

$$0 \to \frac{R}{(I:a)} \xrightarrow{\times a} \frac{R}{I} \to \frac{R}{I+(a)} \to 0.$$

Pasting the composition series of $\frac{R}{(I:a)}$ and $\frac{R}{I+(a)}$, we can get a finite composition series of $\frac{R}{I}$, so $\frac{R}{I}$ is of finite length, contradiction. Thus, I is a prime ideal.

Since I is also maximal ideal. R/I is a field, so R/I is of finite length, contradiction. Thus, $\mathcal{F}$ is empty and R is of finite length. □

Finally, we complete the proof. □

Proposition 0.28

Here are a few proeprty of Artinian rings:

1. Any quotient of an Artinian ring is Artinian.

2. Any localization of an Artinian ring is an Artinian ring.

In module theory, we have the following properties of module of finite length (finitely generated Artinian module over a Noetherian ring):

1. Any submodule of finite length module is of finite length.

2. Any quotient of finite length module is of finite length.

3. Any localization of finite length module is of finite length $(U^{-1}\frac{M_i}{M_{i+1}} \cong U^{-1}R/\mathfrak{m}_i \cong \frac{U^{-1}R}{U^{-1}\mathfrak{m}_i} = 0$ or a simple module).

Now, we put this in a geometric setting:

Theorem 0.29

Let X be an affine variety over an algebraically closed field k. The following are equivalent:

1. X is a finite set of points.
2. $A(X)$ is a finite-dimensional k-vector space, and its dimension is exactly the number of points in X.
3. $A(X)$ is Artinian.

Proof. (1) to (2): Let p_i be the points in X. They are disconnected, so $A(X) = \prod_{p_i \in X} A(p_i) = k^n$ where n is the number of points in X.

(2) to (3): If $A(X)$ is a finite-dimensional k-vector space, then it has a finite composition series as $A(X)$-module, so it's Artinian. Or, any descending chain of ideals of $A(X)$ is also a descending chain of subspaces of $A(X)$, so it must stabilize since the dimension of $A(X)$ is finite.

(3) to (1): If $A(X)$ is Artinian, then $A(X)$ has finitely many prime ideals, which are all maximal. Thus, X has finitely many points. □

Theorem 0.30 (Structure Theorem for Artinian Rings)

Let R be an Artinian ring, which is also module of finite length as R-module. Then we have the isomorphism

$$R \cong \bigoplus_{i=1}^{n} R_{\mathfrak{m}_i}$$

where each $R_{\mathfrak{m}_i}$ is an Artinian local ring.

Proof. The isomorphism we had for R as R-module is actually an isomorphism of rings. The ring structure on $\bigoplus_{i=1}^{n} R_{\mathfrak{m}_i}$ is given by the componentwise multiplication. Beyond that, we need to show the localization of an Artinian ring is Artinian. This is because the localization of an Artinian ring must be Noetherian. Prime ideals of the localized ring correspond to prime ideals of the original ring contained in the maximal ideal we localize at. So in fact, the localized ring has only one prime ideal, which is maximal, so it has Krull dimension 0. Thus, the localized ring is Artinian. □

Corollary 0.31

Let R be a Noetherian ring, and let M be a finitely generated R-module. The following are equivalent:

1. M is of finite length.
2. Some finite product of maximal ideals of R annihilates M. (can be repeated)
3. All primes that contain $\text{Ann}(M)$ are maximal ideals
4. $R/\text{Ann}(M)$ is Artinian.

Proof. (1) to (2): Trivial.
(2) to (3): if a prime ideal $\mathfrak{p}$ contains $\text{Ann}(M) \supseteq \mathfrak{m}_1 \cdots \mathfrak{m}_n$, then $\mathfrak{p}$ contains some $\mathfrak{m}_i$, which is a maximal ideal. Thus, $\mathfrak{p}$ is a maximal ideal.
(3) to (4): Trivial.
(4) to (1): Let $S = \frac{R}{\text{Ann}(M)}$. Since S is Artinian, S is of finite length as S-module. M is finitely generated as R-module, we claim that M is also finitely generated as S-module: $\phi : R^n \twoheadrightarrow M$, we can define a map from S^n to M by sending $r_1 + \text{Ann}(M), \ldots, r_n + \text{Ann}(M)$ to $\phi(r_1, \ldots, r_n)$. This is well-defined since $\phi(\text{Ann}(M)^n) = 0$ and clearly surjective. Thus, M is a quotient of S^n. So M is of finite length as S-module, so it's also of finite length as R-module since any composition factor in S-module is also simple as R-module. $\square$

Corollary 0.32

Let R be a Noetherian ring, and let M be a finitely generated R-module, $I = \text{Ann}(M)$. Then $M_\mathfrak{p}$ is of finite length as $R_\mathfrak{p}$-module if and only if $\mathfrak{p}$ is a minimal prime of I.

This gives us a way to make a module into a module of finite length by localizing at the minimal primes of its annihilator.

Proof. If $\mathfrak{p}$ is a minimal prime of I, then $\mathfrak{p}_\mathfrak{p}$ is the minimal prime of $I_\mathfrak{p}$, which is the nilradical of $R_\mathfrak{p}$. In fact, $\mathfrak{p}_\mathfrak{p}$ is a maximal ideal of $R_\mathfrak{p}$. Thus, $\mathfrak{p}_\mathfrak{p}^n = 0$ for some n and $M_\mathfrak{p}$ is annihilated by $\mathfrak{p}_\mathfrak{p}^n$ (some finite product of maximal ideals), by the corollary above, $M_\mathfrak{p}$ is of finite length as $R_\mathfrak{p}$-module.

Lemma 0.33

If M is finitely generated R-module, then $\operatorname{Ann}(U^{-1}M) \cong U^{-1}\operatorname{Ann}(M)$ as $U^{-1}R$-modules.

This is not ture if M is not finitely generated: take $R = \mathbb{Z}$, $M = \frac{\mathbb{Q}}{\mathbb{Z}}$ and $U = \mathbb{Z}^{\times}$, $\operatorname{Ann}(M) = 0$ but $U^{-1}M = 0$ so $\operatorname{Ann}(U^{-1}M) = \mathbb{Q}$.

With the lemma, we can show the converse. If $M_{\mathfrak{p}}$ is of finite length as $R_{\mathfrak{p}}$-module, then $R_{\mathfrak{p}}/\operatorname{Ann}(M_{\mathfrak{p}})$ is Artinian. $\operatorname{Ann}(M_{\mathfrak{p}}) = I_{\mathfrak{p}}$, so $R_{\mathfrak{p}}/I_{\mathfrak{p}}$ is Artinian. Moreover, $R_{\mathfrak{p}}/I_{\mathfrak{p}}$ is a local ring, so it has only one prime/maximal ideal. So $\mathfrak{p}_{\mathfrak{p}}$ is the unique prime ideal of $R_{\mathfrak{p}}$ containing $I_{\mathfrak{p}}$. Thus, $\mathfrak{p}$ is the unique prime ideal of R over I contained in $\mathfrak{p}$, so $\mathfrak{p}$ is a minimal prime of I. $\square$

Corollary 0.34

Let I be an ideal of a Noetherian ring R and $\mathfrak{p}$ is a prime containing I. Then the following are equivalent:

1. $\mathfrak{p}$ is a minimal prime of I.
2. $\frac{R_{\mathfrak{p}}}{I_{\mathfrak{p}}}$ is Artinian.
3. There eixsts n such that $\mathfrak{p}_{\mathfrak{p}}^n \subseteq I_{\mathfrak{p}}$.

The idea of the proof is the same as the previous corollary.